

Analyzing Cardiovascular Diseases through Data Mining Techniques

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Data Mining Final Project

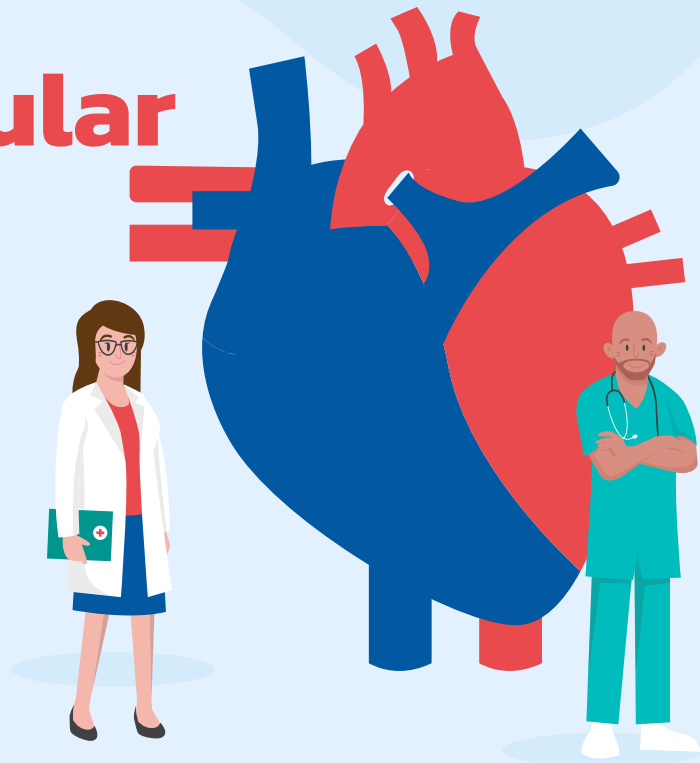


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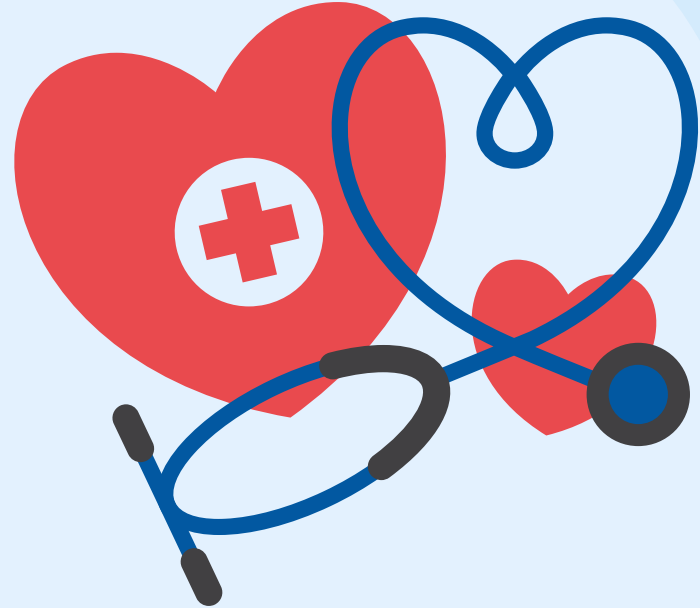
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01

About CVDs

Background and Motivation
Problem



What are CVDs?

- **Cardiovascular diseases** (CVDs) are the leading cause of death globally, taking an estimated 17.9 million lives each year. (WHO)
- They are a group of disorders of the heart and blood vessels and include coronary heart disease, cerebrovascular disease, rheumatic heart disease and other conditions. (WHO)
- CVDs are one of the top 10 causes of death here in **Taiwan**.
- According to the Ministry of Health and Welfare, it is also among the top three leading causes of death.

What are CVDs?

- Having unhealthy diets, physical inactivity, the usage of tobacco, and excessive alcohol consumption constitute the primary behavioral risk factors for heart diseases and stroke.

表 1 十大死因死亡人數及死亡率

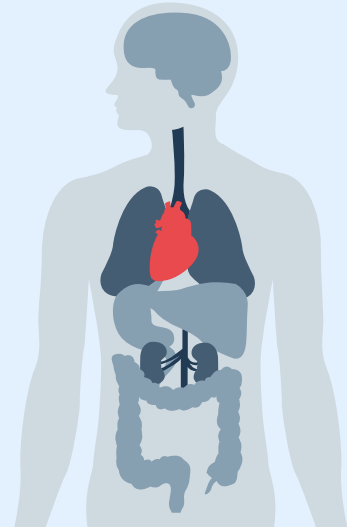
順位	所有死亡原因	死亡人數(人)		死亡率 (每十萬人口)			標準化死亡率 (每十萬人口)		
		111年	較上年 增減%	110年 順位	111年	較上年 增減%	順位	111年	較上年 增減%
	所有死亡原因	208,438	13.2		893.8	13.9		443.9	9.5
1	癌症	51,927	0.5	1	222.7	1.2	1	116.0	-1.8
2	心臟疾病 (高血壓性疾病除外)	23,668	8.3	2	101.5	9.0	2	47.8	4.8
3	嚴重特殊傳染性肺炎 (COVID-19)	14,667	1,536.9	19	62.9	1,547.4	3	28.6	1,365.5
4	肺炎	14,320	5.7	3	61.4	6.4	4	26.3	3.8
5	腦血管疾病	12,416	1.9	4	53.2	2.6	5	25.1	-0.3
6	糖尿病	12,289	7.3	5	52.7	8.0	6	24.7	3.7
7	高血壓性疾病	8,720	10.6	6	37.4	11.3	8	16.3	7.2
8	事故傷害	6,953	2.6	7	29.8	3.3	7	20.0	0.2
9	慢性下呼吸道疾病	6,494	4.1	8	27.8	4.8	10	11.8	1.6
10	腎炎、腎病症候群及腎病變	5,813	6.3	9	24.9	6.9	11	11.3	3.5

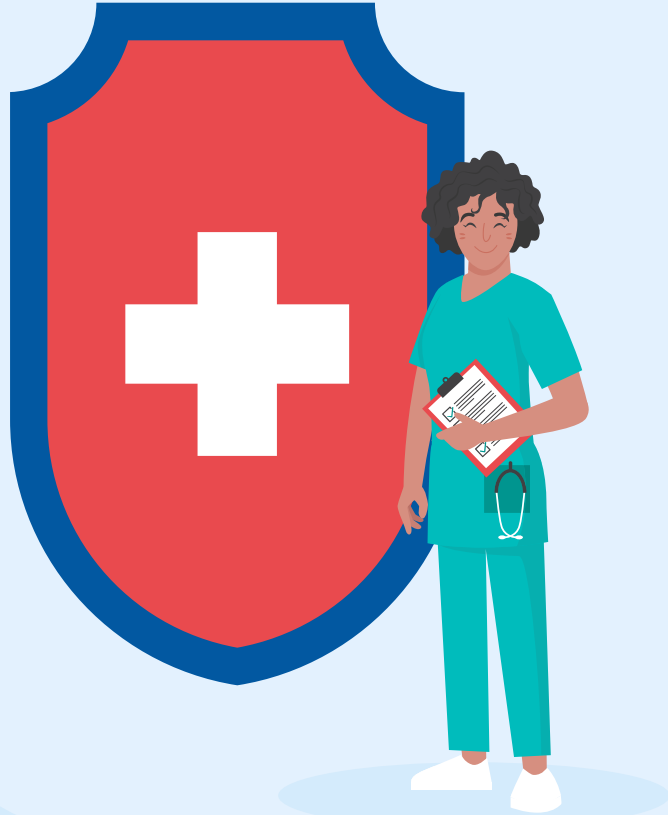
Background & Motivation

- Because there are no obvious symptoms regarding the blood vessels, the first indication of it may present itself as a heart attack or stroke, which may cause great damage to our body.
- Therefore, analyzing cardiovascular diseases through data mining techniques may offer us more understanding of the relations among those intermediate risk factors.

Problem

- There are numerous potential factors that can lead to the CVDs.
- The following research addresses '**what kinds of factors cause CVDs significantly and their relationships by constructing classification models**'.





02

About Data

Data Description
The Importance of Data

Data Description

- The used dataset comes from [Cardiovascular Disease dataset \(kaggle.com\)](https://www.kaggle.com/datasets/ahmedmohamednassef/coronary-artery-disease-dataset)
- The dataset uploaded to Kaggle five years ago.
- There is no licenses, and all values were collected at the moment of medical examination.
- However, it is worth noting that a few papers use the same dataset, which can demonstrate its effectiveness.

The Importance of Data

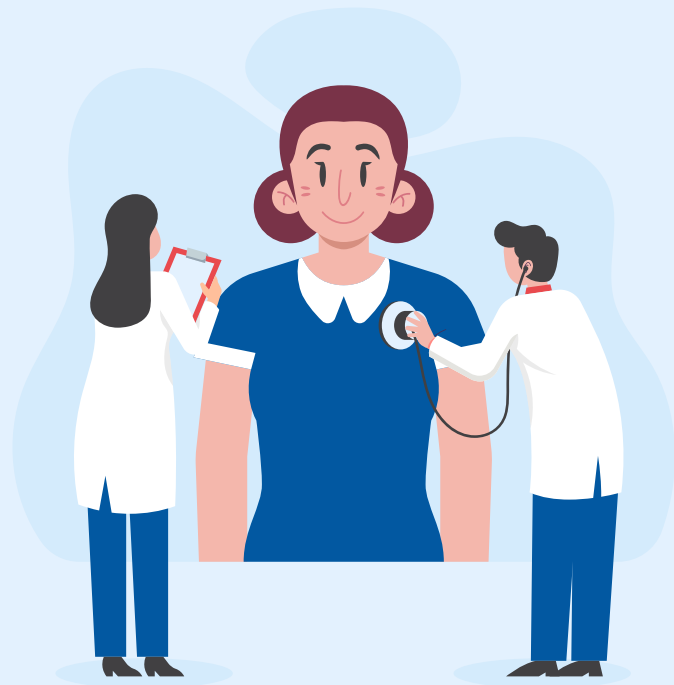
- The dataset contains exactly **70,000 records** with **12 CVD-related attributes** except for patients' IDs.
- There are 3 types of input features:
 - **Objective** (factual information)
 - **Examination** (results of medical examination)
 - **Subjective** (information given by the patient)

Detailed information of cardiovascular dataset				
	Column Name	Input Feature	Explanation	Data Type
Meaningless	id		Id number	integer
	age	Objective Feature	Age	integer (days)
	height	Objective Feature	Height	integer (cm)
	weight	Objective Feature	Weight	integer (kg)
Nominal Scale	gender	Objective Feature	Gender	categorical code (1-women; 2-men)
	ap_hi	Examination Feature	Systolic blood pressure	integer
	ap_lo	Examination Feature	Diastolic blood pressure	integer
Ordinal Scale	cholesterol	Examination Feature	Cholesterol	1: normal; 2: above normal; 3: well above normal
Ordinal Scale	gluc	Examination Feature	Glucose	1: normal; 2: above normal; 3: well above normal
Nominal Scale	smoke	Subjective Feature	Smoking	binary
Nominal Scale	alco	Subjective Feature	Alcohol intake	binary
Nominal Scale	active	Subjective Feature	Physical activity	binary
Target Value	cardio	Target Variable	Presence or absence of cardiovascular disease	binary

03

Data Analysis

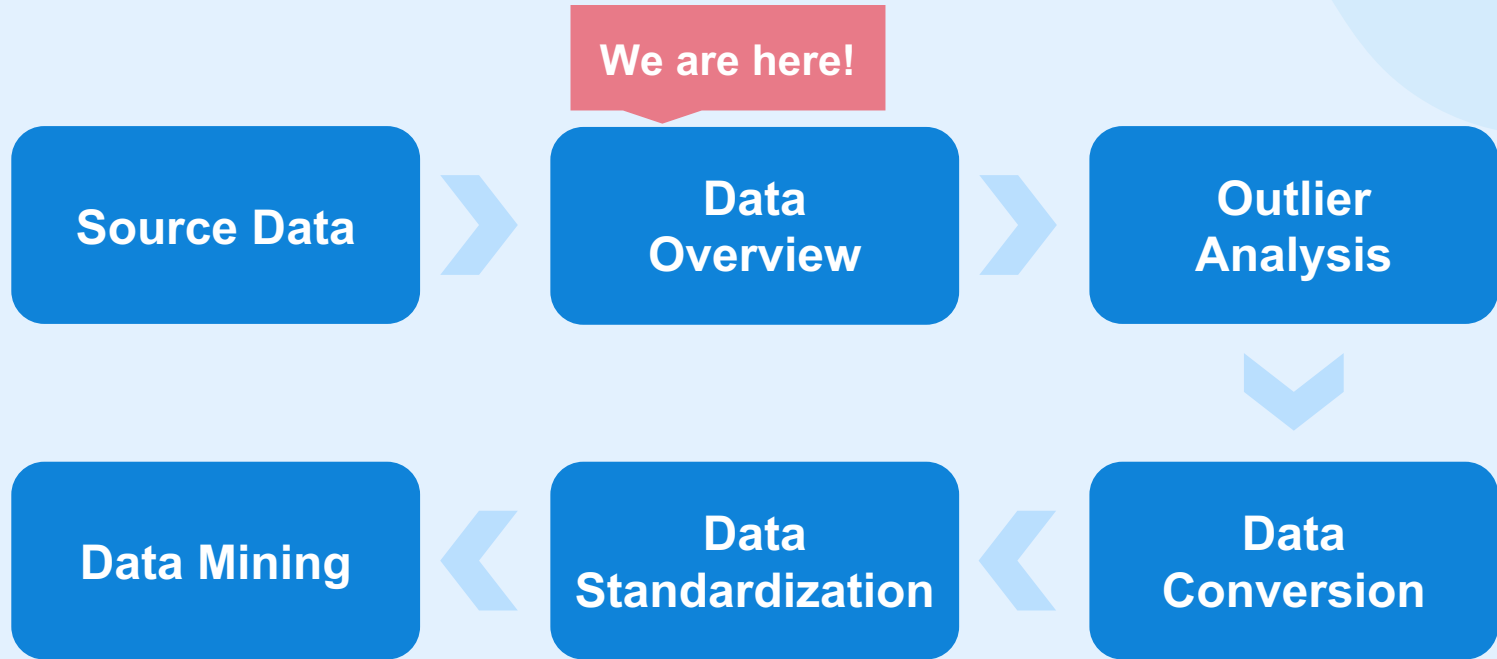
Data Preprocessing



Data Preprocessing

- Given that there is **no missing value** in the dataset, the process of missing value can be ignored.
- However, **outlier analysis** and **data conversion/standardization** are still essential.
- Thus, I will utilize **SPSS statistics and Colab** to tackle the issues above.

Data Preprocessing



Data Preprocessing – data overview

- From the SPSS report below, we could know that the dataset contains 70,000 records and no missing value.

		Statistics				
		age	height	weight	ap_hi	ap_lo
N	Valid	70000	70000	70000	70000	70000
	Missing	0	0	0	0	0
Mean		19468.87	164.36	74.2057	128.82	96.63
Std. Error of Mean		9.325	.031	.05441	.582	.712
Median		19703.00	165.00	72.0000	120.00	80.00
Mode		18236 ^a	165	65.00	120	80
Std. Deviation		2467.252	8.210	14.39576	154.011	188.473
Variance		6087330.790	67.406	207.238	23719.517	35521.895
Skewness		-.307	-.642	1.012	85.296	32.114
Std. Error of Skewness		.009	.009	.009	.009	.009
Kurtosis		-.823	7.944	2.587	7580.075	1425.915
Std. Error of Kurtosis		.019	.019	.019	.019	.019
Range		12915	195	190.00	16170	11070
Minimum		10798	55	10.00	-150	-70
Maximum		23713	250	200.00	16020	11000

Statistics		
		N
		Valid
		Missing
gender	70000	0
cholesterol	70000	0
gluc	70000	0
smoke	70000	0
alco	70000	0
active	70000	0
cardio	70000	0

Data Preprocessing – data overview

- By examining the descriptive statistics of numerical data, there are some values beyond the normal situation. For example, the lowest diastolic blood pressure of ap_lo is -70.
- Thus, deleting/adjusting some abnormal data is necessary.

		Descriptive Statistics					Valid N (listwise)
		age	height	weight	ap_hi	ap_lo	
N	Statistic	70000	70000	70000	70000	70000	70000
Minimum	Statistic	10798	55	10.00	-150	-70	
Maximum	Statistic	23713	250	200.00	16020	11000	
Sum	Statistic	1362820607	11505146	5194398.30	9017210	6764129	
Mean	Statistic	19468.87	164.36	74.2057	128.82	96.63	
Std. Deviation	Statistic	2467.252	8.210	14.39576	154.011	188.473	
Variance	Statistic	6087330.790	67.406	207.238	23719.517	35521.895	
Skewness	Statistic	-.307	-.642	1.012	85.296	32.114	
	Std. Error	.009	.009	.009	.009	.009	
Kurtosis	Statistic	-.823	7.944	2.587	7580.075	1425.915	
	Std. Error	.019	.019	.019	.019	.019	

Data Preprocessing – data overview

- Note that the normal systolic blood pressure should be lower than 120 mmHg and diastolic blood pressure should below 80 mmHg.
- If the value is greater than the above, this indicates that there is some high blood pressure crisis. (American Heart Association)

BLOOD PRESSURE CATEGORY	SYSTOLIC mm Hg (upper number)	and/or	DIASTOLIC mm Hg (lower number)
NORMAL	LESS THAN 120	and	LESS THAN 80
ELEVATED	120 – 129	and	LESS THAN 80
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1	130 – 139	or	80 – 89
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2	140 OR HIGHER	or	90 OR HIGHER
<u>HYPERTENSIVE CRISIS</u> (consult your doctor immediately)	HIGHER THAN 180	and/or	HIGHER THAN 120

Data Preprocessing – data overview

- For the ordinal and nominal data, the frequency tables are shown below:

gender

	N	%
1	45530	65.0%
2	24470	35.0%

smoke

	N	%
0	63831	91.2%
1	6169	8.8%

gluc

	N	%
1	59479	85.0%
2	5190	7.4%
3	5331	7.6%

cholesterol

	N	%
1	52385	74.8%
2	9549	13.6%
3	8066	11.5%

alco

	N	%
0	66236	94.6%
1	3764	5.4%

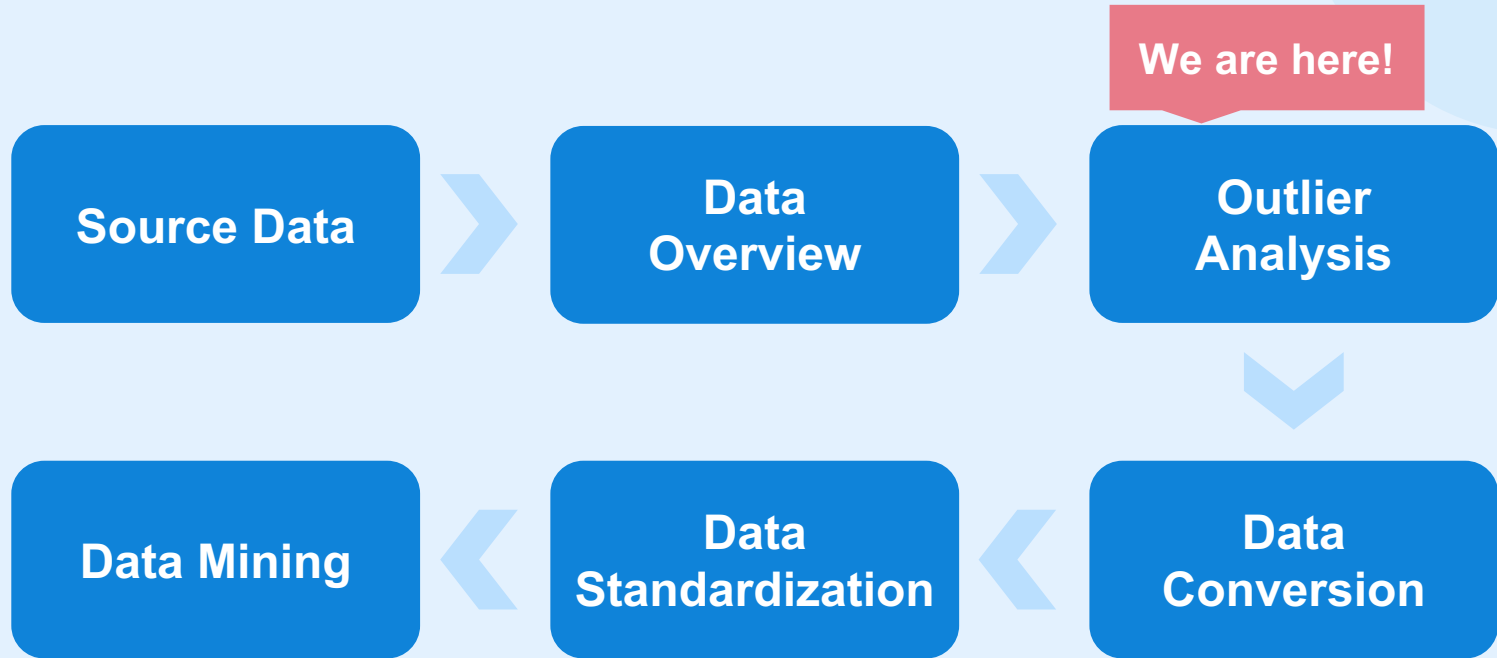
active

	N	%
0	13739	19.6%
1	56261	80.4%

cardio

	N	%
0	35021	50.0%
1	34979	50.0%

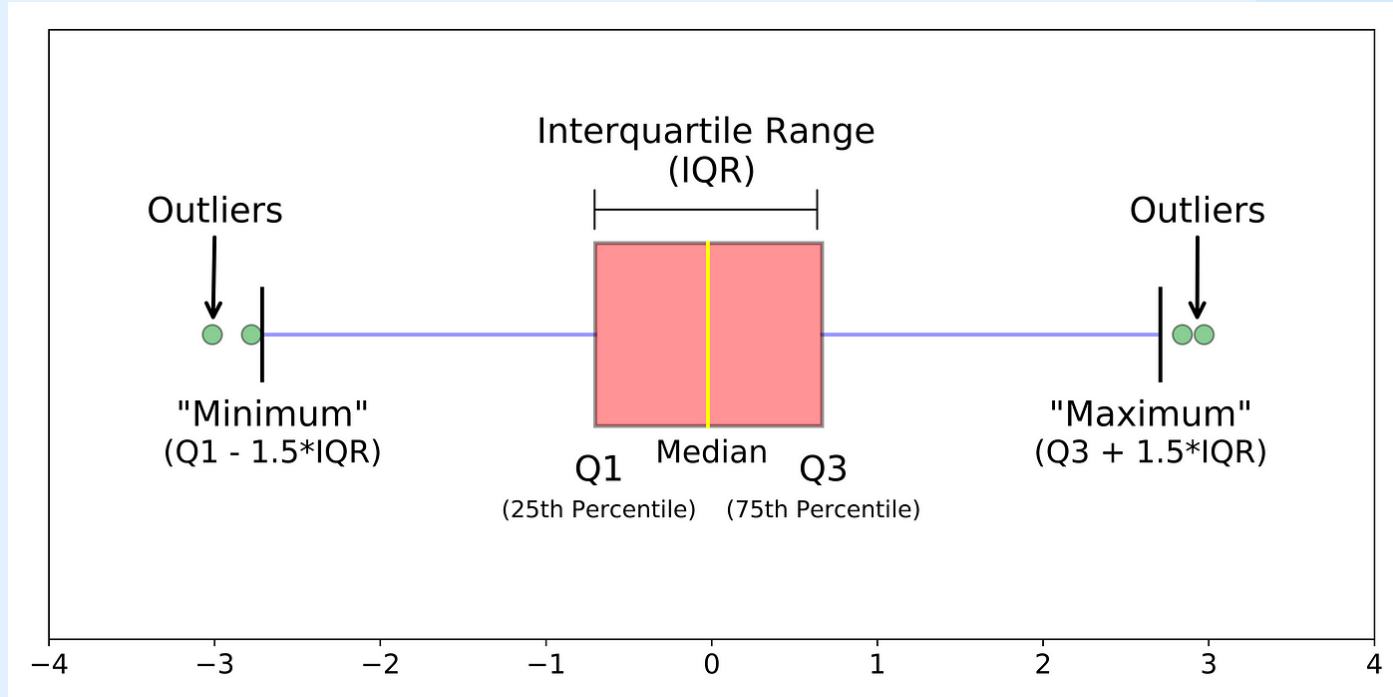
Data Preprocessing



Data Preprocessing – outlier analysis

- When executing outlier analysis, we can use boxplot and histogram to examine the possible outliers.
- Most of the time, the outliers can be defined as
$$\begin{cases} \text{Outlier} = Q_1 - 1.5IQR \\ \text{or} \\ \text{Outlier} = Q_3 + 1.5IQR \end{cases}$$
- At the same time, we can use **log()** to deal with the skewness problem and also reduce the effect of outliers, especially right-skewed.
- Additionally, we must be careful to exclude those abnormal data, e.g. the negative value in systolic/diastolic blood pressure.

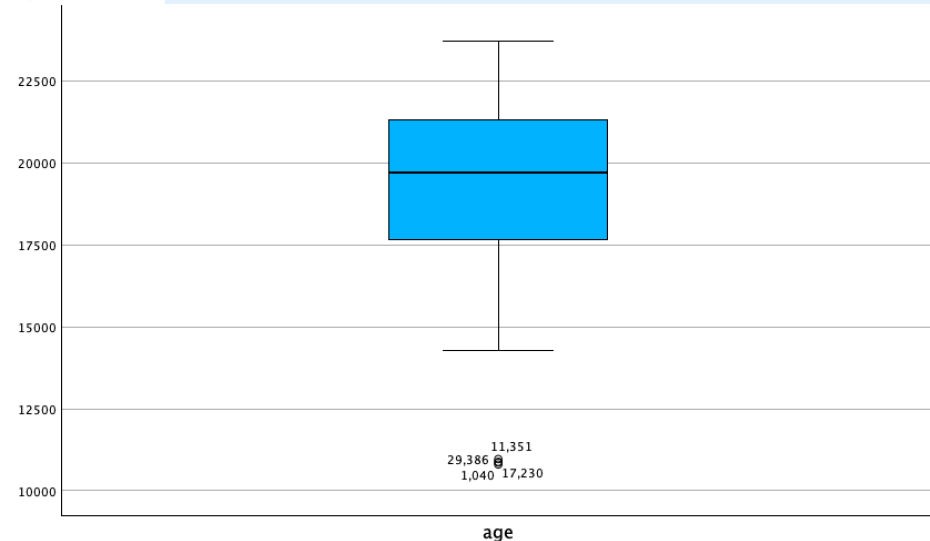
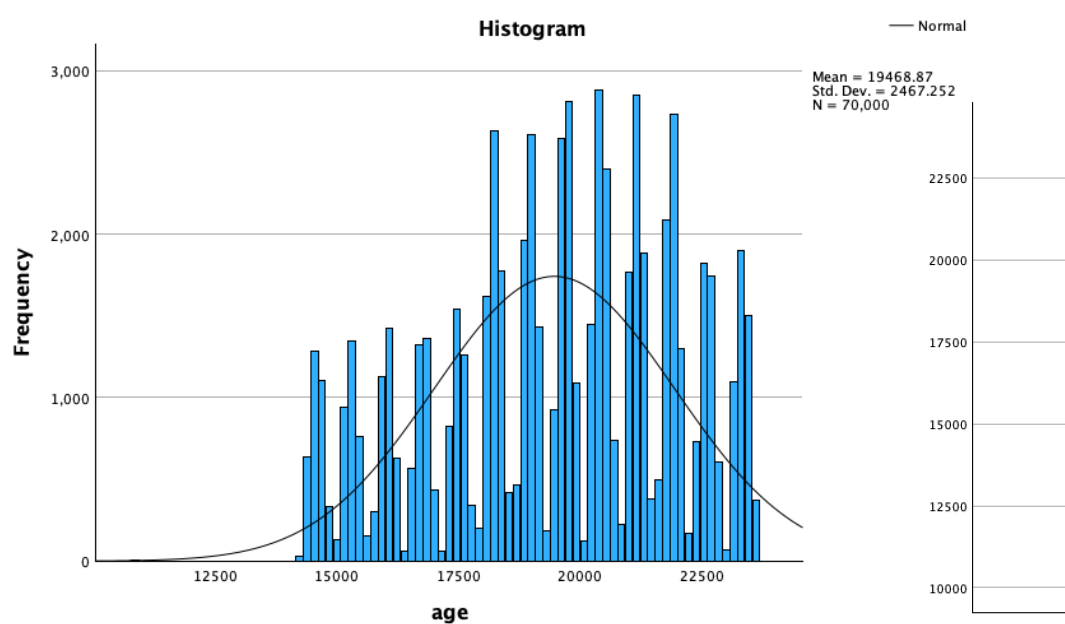
Data Preprocessing – outlier analysis



[Removing outliers from data using Python and Pandas | by Graham Harrison | Analytics Vidhya | Medium](#)

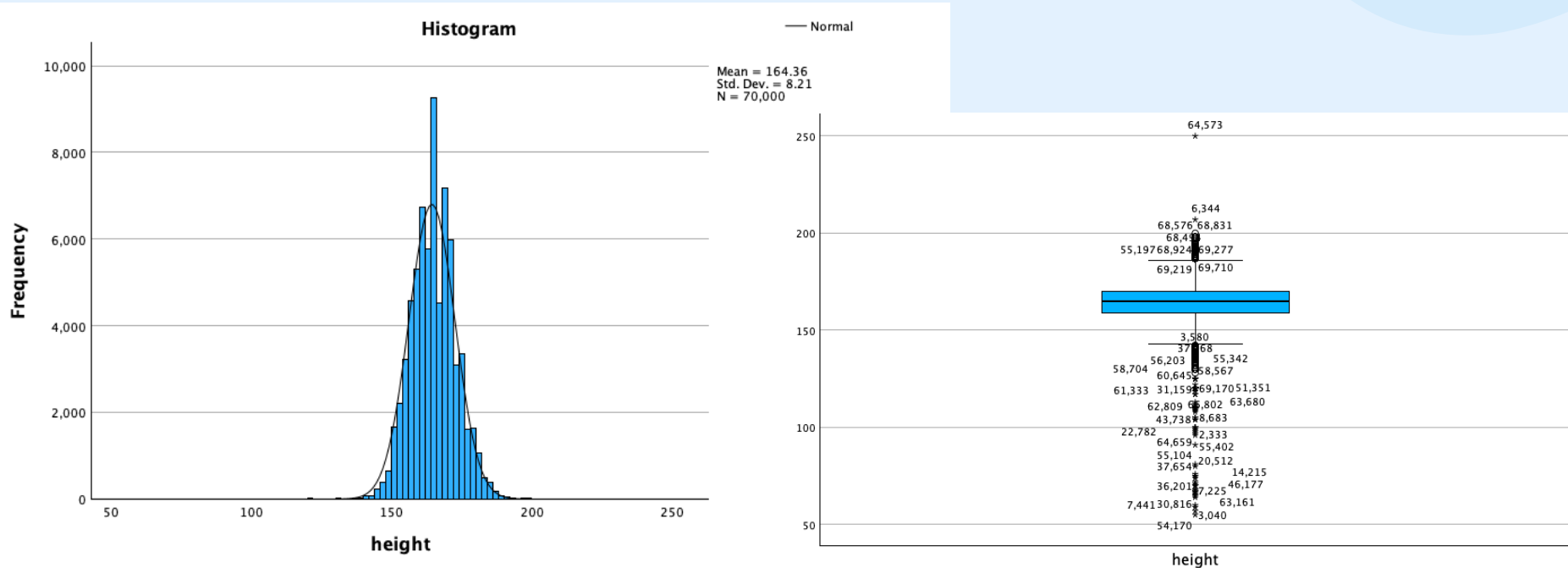
Data Preprocessing – outlier analysis

- By examining the histogram and boxplot of the age attribute, there are some possible outliers, which far away from the lower extreme.
- The above condition will be improved at conversion phase.



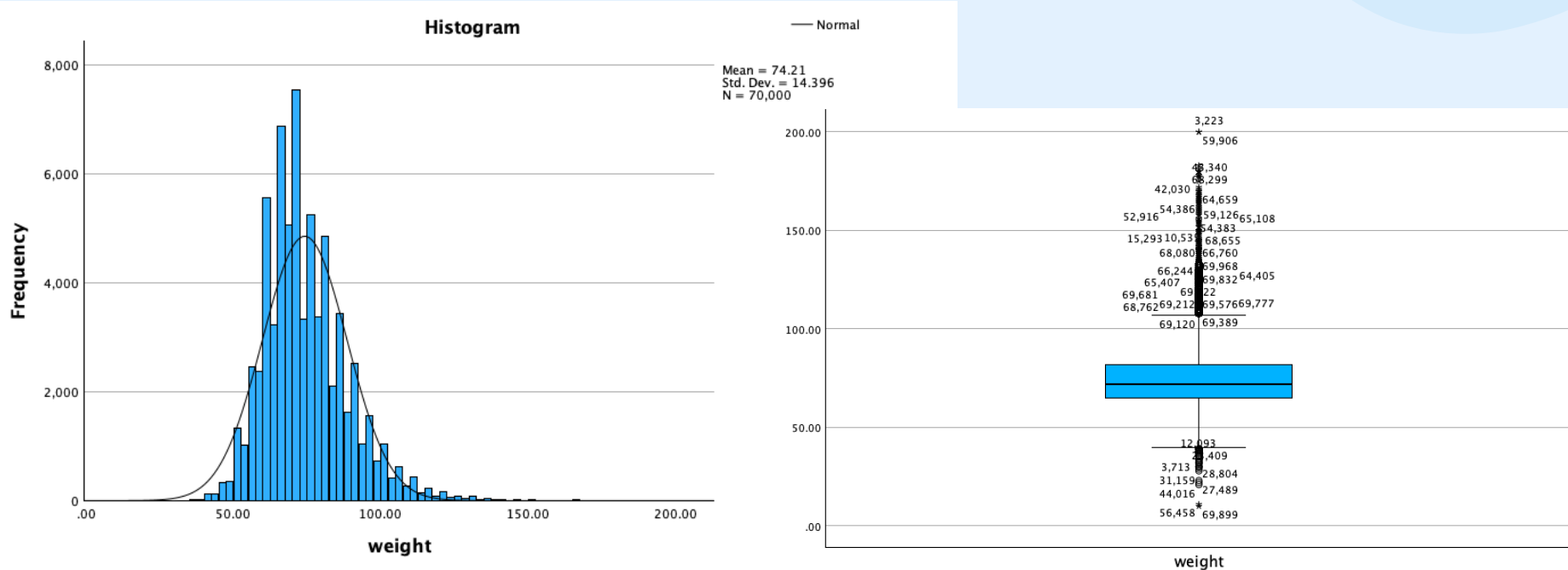
Data Preprocessing – outlier analysis

- By examining the histogram and boxplot of the height attribute, there are a lot outliers (both upper and lower extreme).



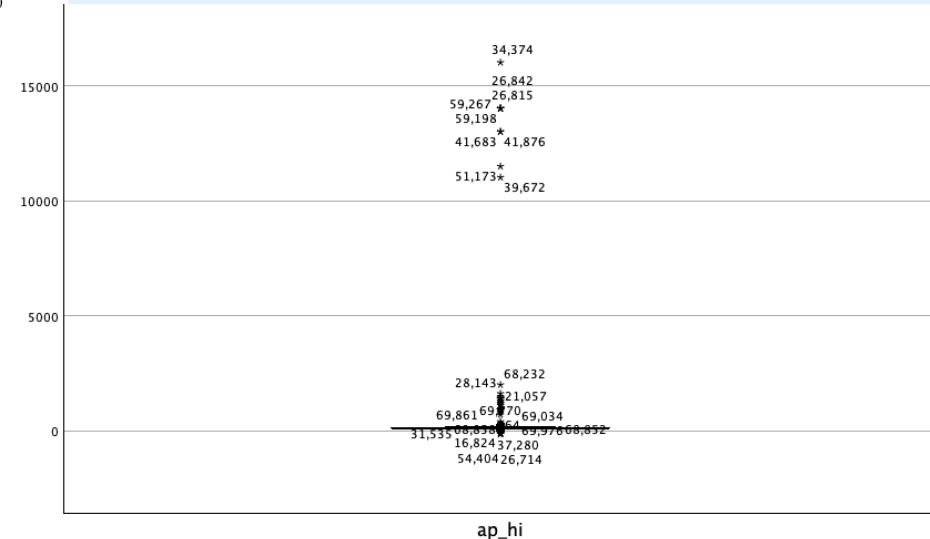
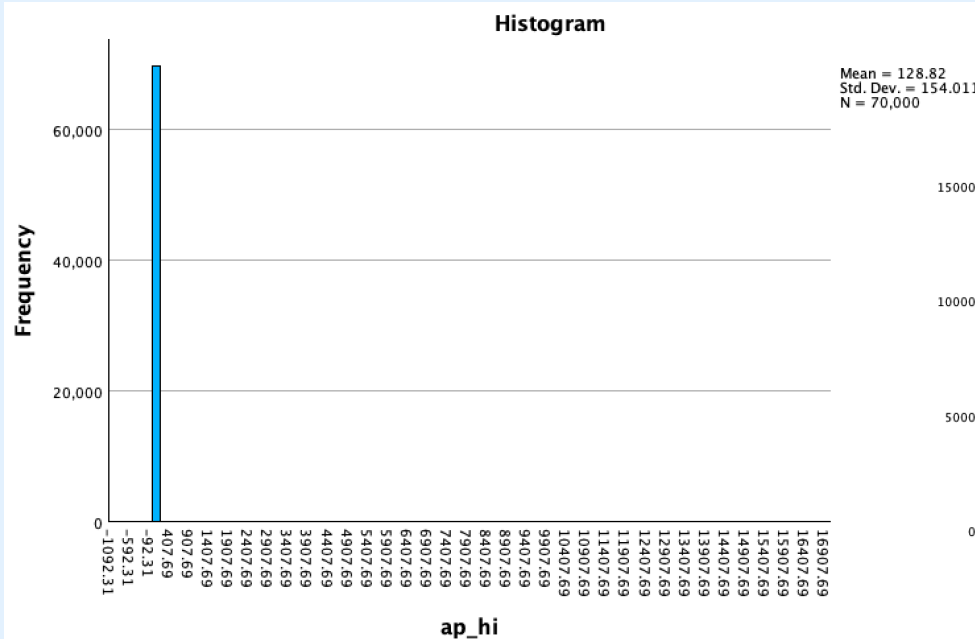
Data Preprocessing – outlier analysis

- By examining the histogram and boxplot of the weight attribute, there are lots of outliers (same as the height attribute).



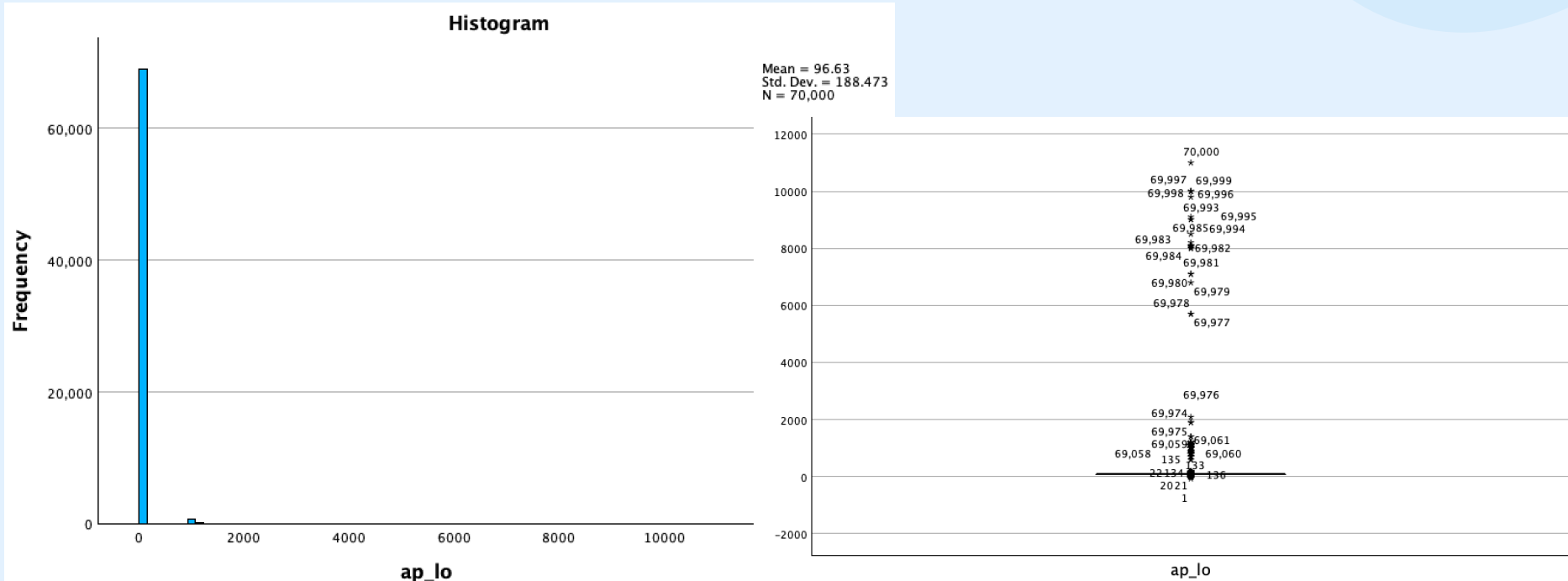
Data Preprocessing – outlier analysis

- By examining the histogram and boxplot of the systolic blood pressure attribute, there are numerous 'extreme' that cause the graph cannot be shown normally.



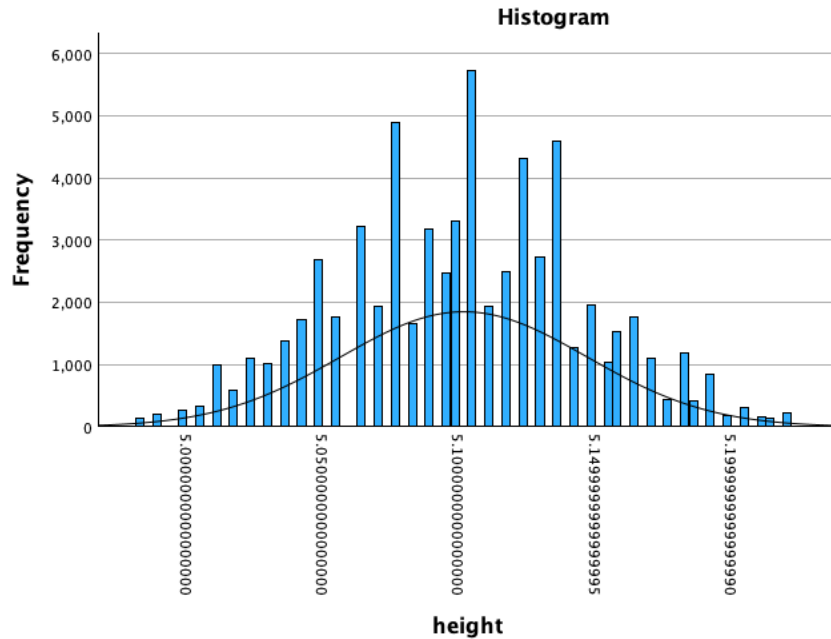
Data Preprocessing – outlier analysis

- By examining the histogram and boxplot of the diastolic blood pressure attribute, the situation is identical to the ap_hi attribute.



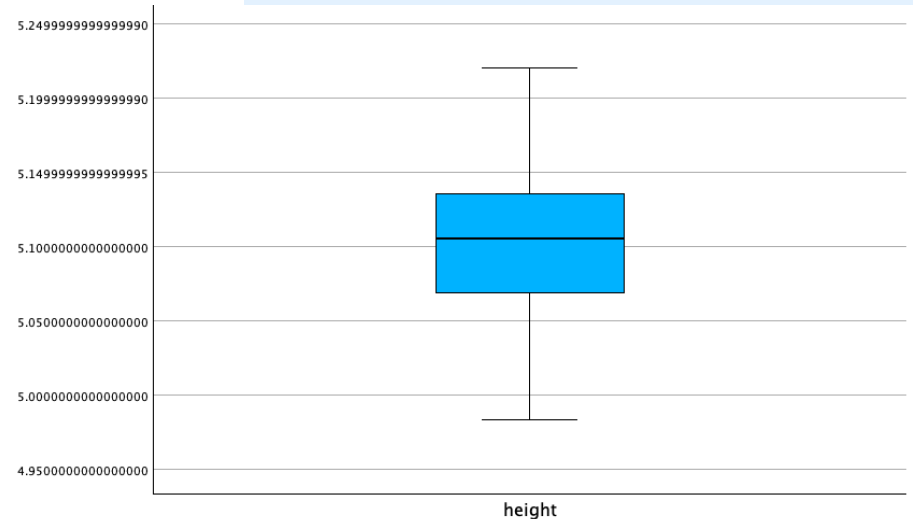
Data Preprocessing – outlier analysis

- After adjusting, there are no outliers in the height attribute and the distribution is bell-shaped.



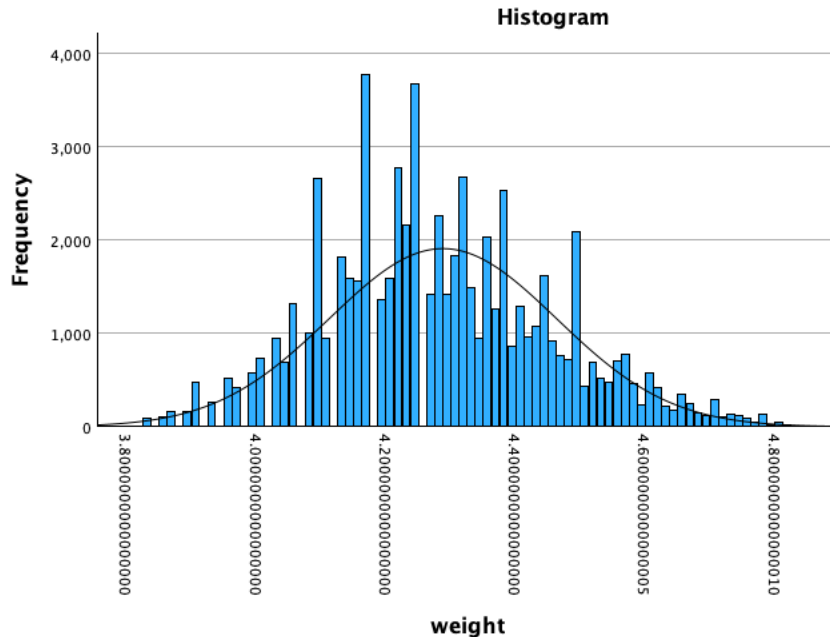
— Normal

Mean = 5.1017360165255040
Std. Dev. = .0452683443713598
N = 67,118



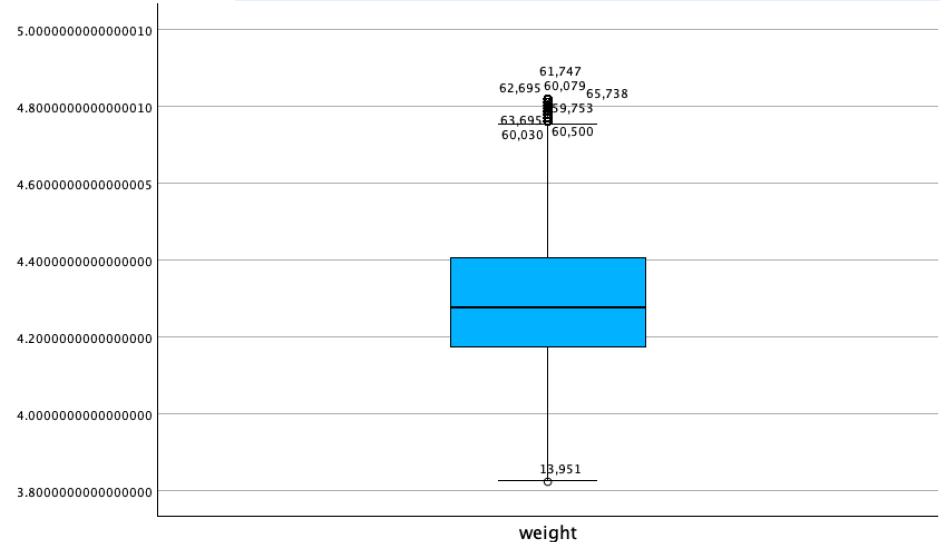
Data Preprocessing – outlier analysis

- After adjusting, there are still some values outside the upper extreme in the weight attribute, however, the distribution is bell-shaped.



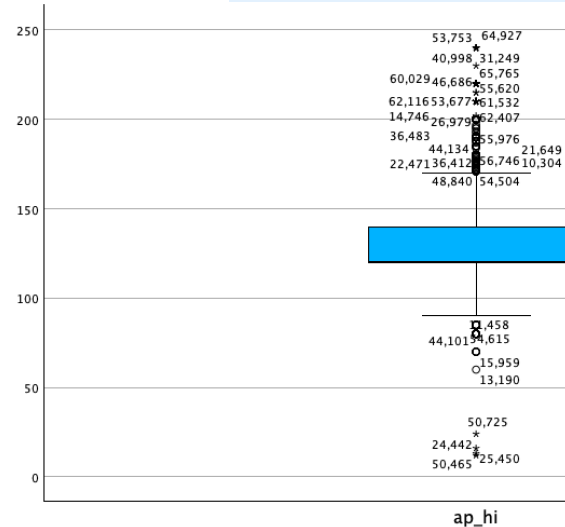
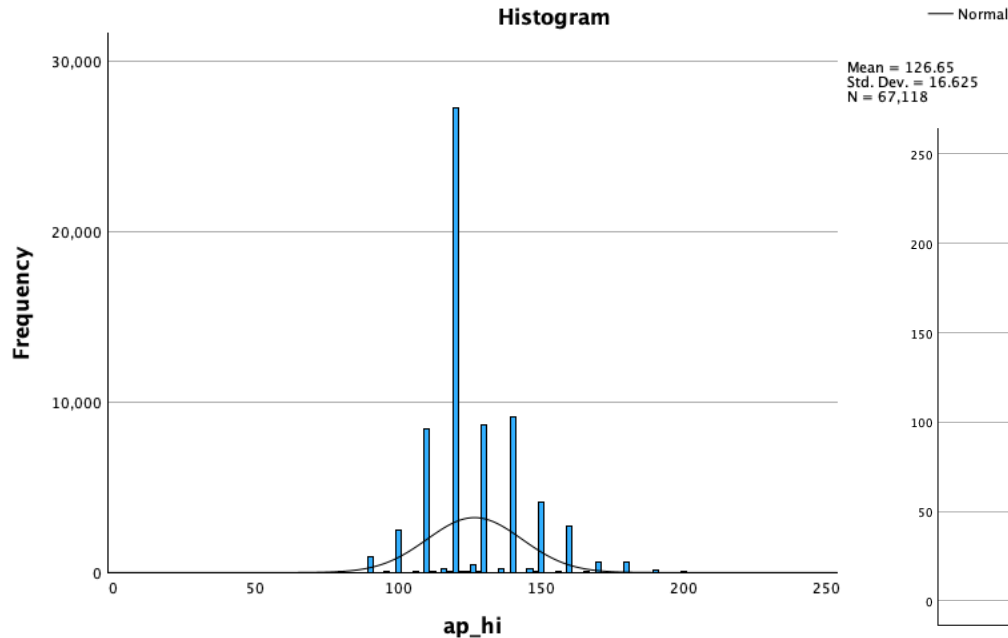
— Normal

Mean = 4.2877388336488025
Std. Dev. = .1753580058096903
N = 67,118



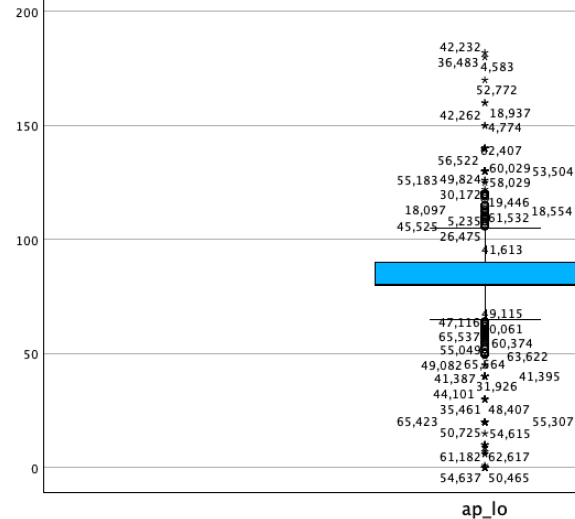
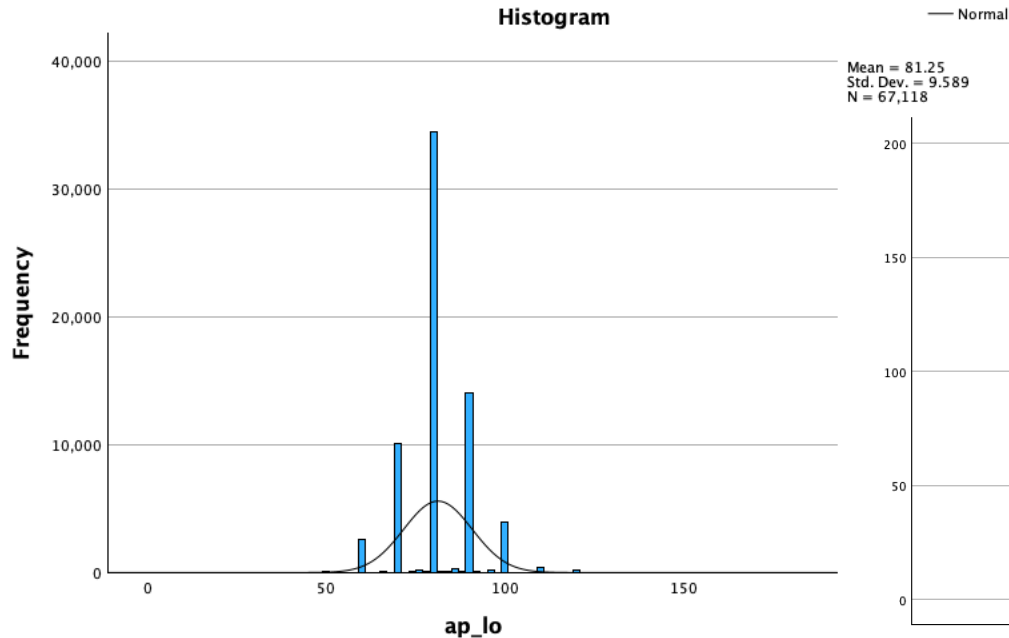
Data Preprocessing – outlier analysis

- After adjusting, there are still some possible outliers. However, the value of ap_hi is much more regular.



Data Preprocessing – outlier analysis

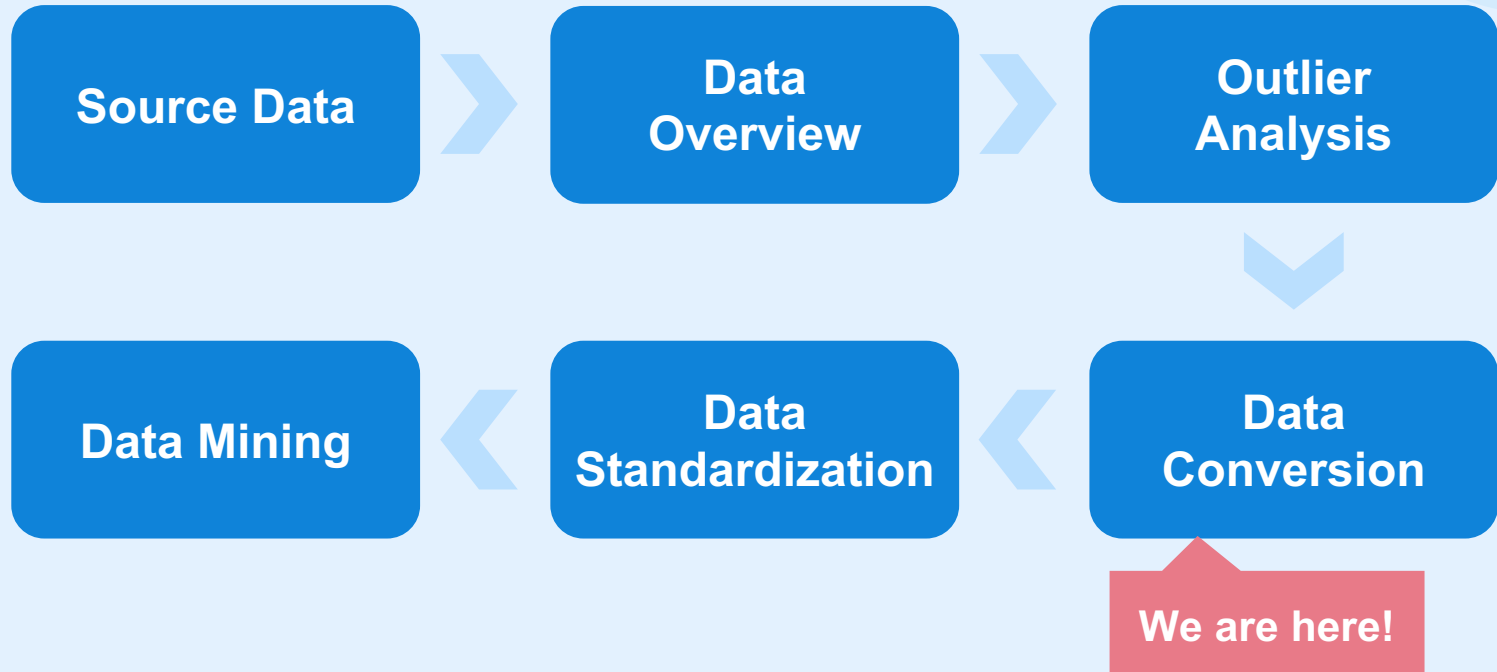
- After adjusting, the condition is similar to the ap_hi attribute.



Data Preprocessing – outlier analysis

- In conclusion, for numeric type of data, we first look at their original distribution and then examine the reasonableness either utilize $\log()$ to adjust the outliers and skewness or remove those insane data directly.

Data Preprocessing



Data Preprocessing – data conversion

- For better understanding of the age attribute, the unit of '**days**' should be converted to '**years**' although it will become slightly inaccurate. However, the slight change could be ignored because a little shift will not cause significant influence in detecting CVDs.
- At the same time, the height and weight attribute seem to be combined together to generate a new attribute 'BMI'.
- Note that **the body mass index** (BMI) is a measurement that evaluate personal height and weight as a whole.

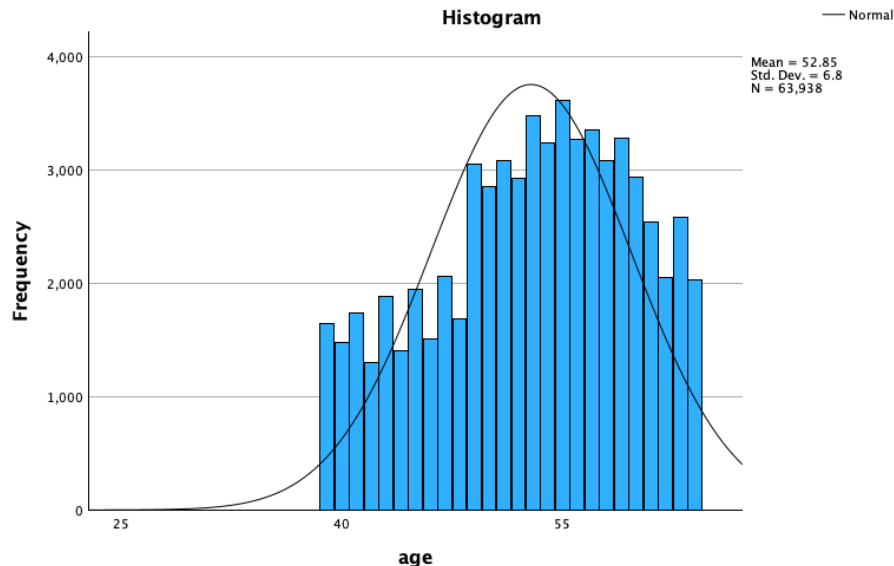
Data Preprocessing – data conversion

- For the following descriptives and histogram of the age attribute, the minimum age is 29 and the maximum is 64.
- The skewness and kurtosis values are acceptable.

Descriptives

		Statistic	Std. Error
age	Mean	52.85	.027
	95% Confidence Interval for Mean	Lower Bound	52.80
		Upper Bound	52.91
	5% Trimmed Mean	53.00	
	Median	53.00	
	Variance	46.246	
	Std. Deviation	6.800	
	Minimum	29	
	Maximum	64	
	Range	35	
	Interquartile Range	10	
	Skewness	-.303	.010
	Kurtosis	-.837	.019

Histogram



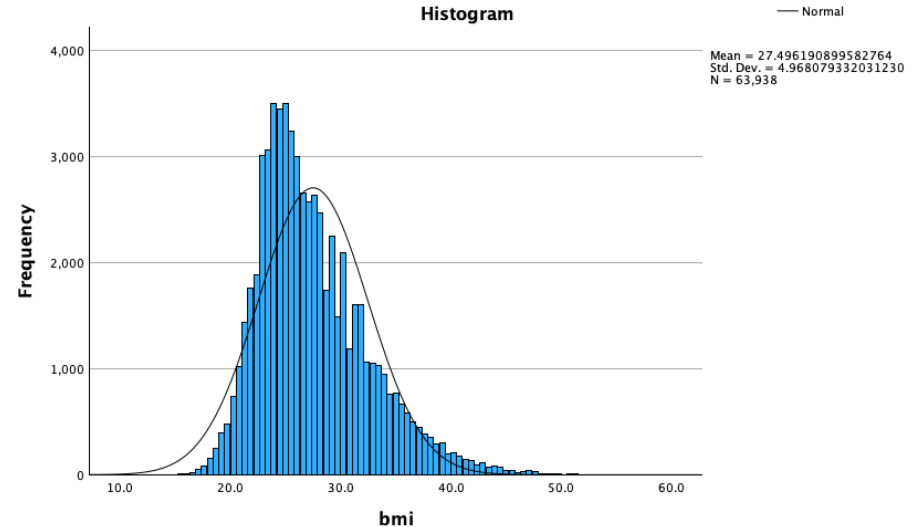
Data Preprocessing – data conversion

- For the following descriptives and histogram of the BMI attribute, the minimum age is 14.6 and the maximum is 56.2.
- The skewness and kurtosis values are acceptable.

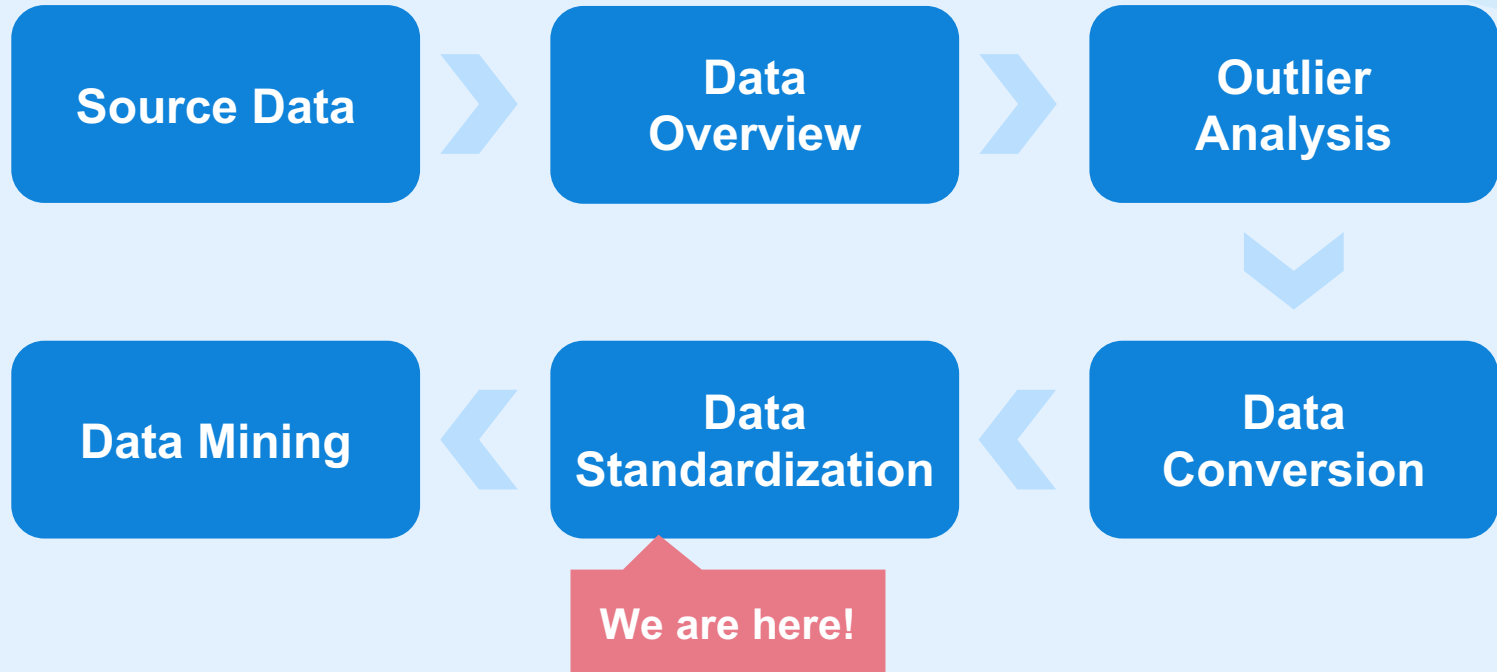
Descriptives

		Statistic	Std. Error
bmi	Mean	27.4961909	.019647577
	95% Confidence Interval for Mean	Lower Bound	27.4576816
		Upper Bound	27.5347002
	5% Trimmed Mean	27.2138084	
	Median	26.5432099	
	Variance	24.682	
	Std. Deviation	4.96807933	
	Minimum	14.6092038	
	Maximum	56.2957403	
	Range	41.6865365	
	Interquartile Range	6.49191441	
	Skewness	.936	.010
	Kurtosis	1.049	.019

Histogram



Data Preprocessing

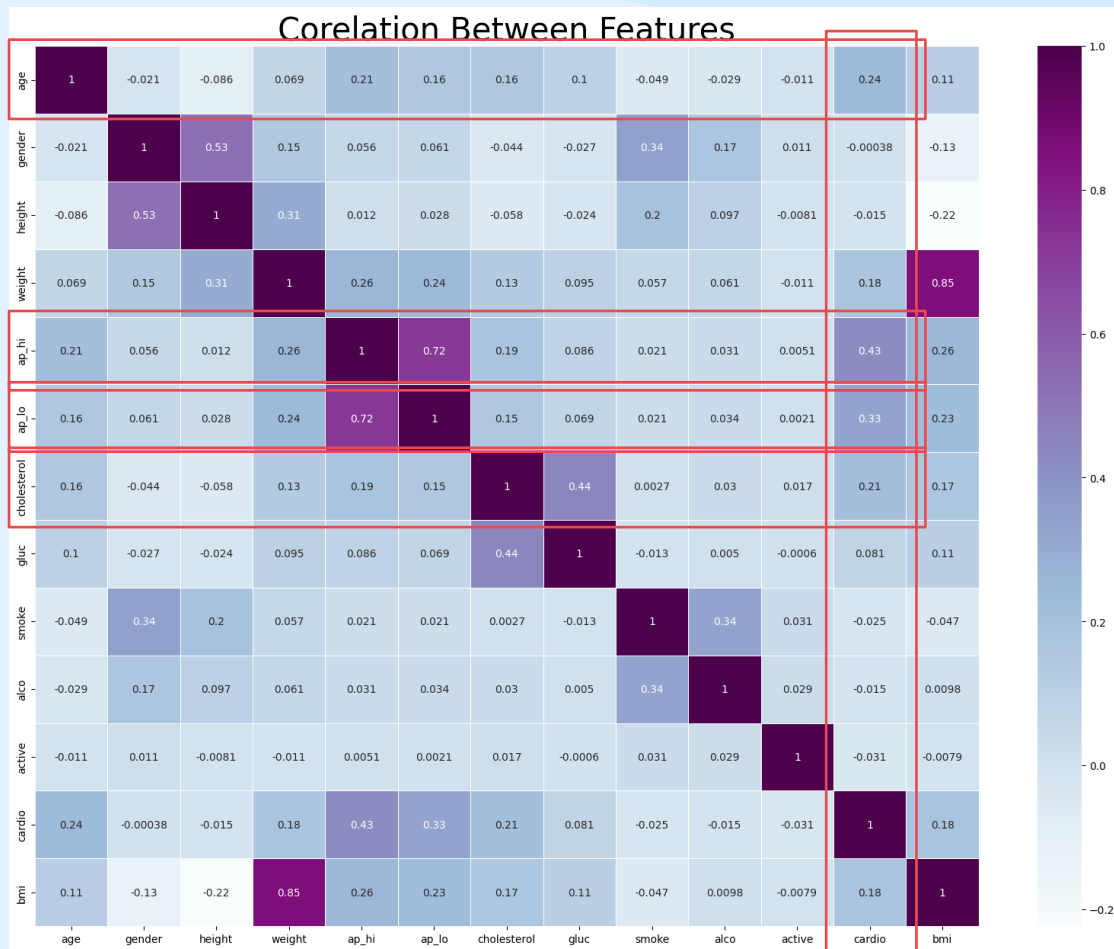


Data Preprocessing – data standardization

- Before further processing the data, heat map is a great tool to visualize the correlation between the target value and the input values.
- By using it, the final input features can be determined, which have a relative strong correlation to the target value.

Data Preprocessing – data standardization

- The following heat map is generated on Colab.



Data Preprocessing – data standardization

- From the heat map (previous slide), we can know that the 'age', 'ap_hi', 'ap_lo', and 'cholesterol' attribute have relative positive correlation. (Though none of them exceed 0.5)
- Therefore, the above selected attributes will further proceed.

Rule of Thumb for Interpreting the Size of a Correlation Coefficient⁴

Size of Correlation	Interpretation
.90 to 1.00 (–.90 to –1.00)	Very high positive (negative) correlation
.70 to .90 (–.70 to –.90)	High positive (negative) correlation
.50 to .70 (–.50 to –.70)	Moderate positive (negative) correlation
.30 to .50 (–.30 to –.50)	Low positive (negative) correlation
.00 to .30 (.00 to –.30)	negligible correlation

[A guide to appropriate use of Correlation coefficient in medical research - PMC \(nih.gov\)](#)

Data Preprocessing – data standardization

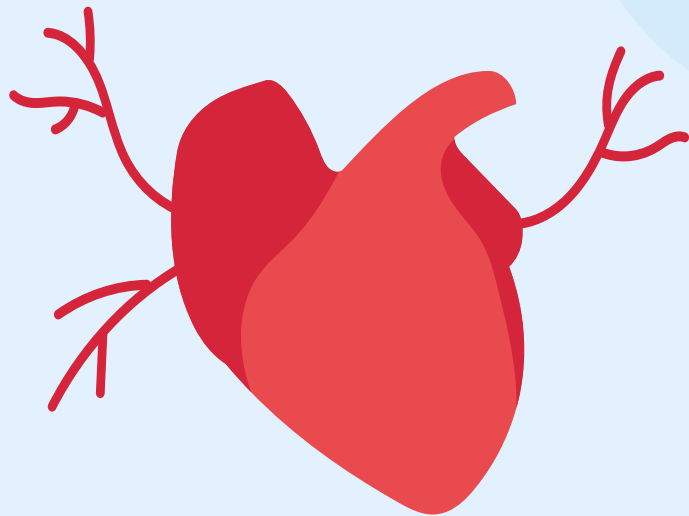
- Finally, **StandardScaler()** method is applied, which is used for standardizing features by removing the mean and scaling to unit variance. In other words, it is to transform the features such that they have a mean of 0 and a standard deviation of 1.



04

Model and The DM algorithms

Input/Output Features
Used DM Algorithms



Model and The DM algorithms

- After finishing the data preprocessing, the next step is to train the predictive model.
- For the input feature, the '**age**', '**ap_hi**', '**ap_lo**', and '**cholesterol**' attribute are chosen.
- The used data mining algorithms are the following:
k-NN, Random Forest, SVM & XGBoost, respectively.
- All the training data is split into (x_train, x_test, y_train, y_test) and under the condition of 75% training size and 25% testing size.
- Also, the random seed is 42.

Model and The DM algorithms – kNN

- The k-NN is simple and intuitive and suitable for medium size datasets. Although the CVDs dataset is quite large, it is still a good way to build model.

```
params = {'n_neighbors':list(range(0, 50)),  
          'weights':['uniform', 'distance'],  
          'p':[1,2]}
```



```
knn = KNeighborsClassifier()  
knn_grid_cv = GridSearchCV(knn, param_grid=params, cv=10)  
knn_grid_cv.fit(X_train, y_train)  
print("Best Hyper Parameters:\n",knn_grid_cv.best_params_)
```



```
Best Hyper Parameters:  
{'n_neighbors': 42, 'p': 1, 'weights': 'uniform'}
```

Result

Model and The DM algorithms – Random Forest

- The random forest is robust and effective for classification and it can handle large datasets efficiently.

```
params = {  
    'n_estimators': [10, 50, 100, 150, 200, 300, 400,  
500],  
    'max_depth' : [10,20,30,40,50],  
    'criterion' : ['entropy','gini']  
}
```

```
rfc_gridcv = RandomForestClassifier(random_state=42)  
rfc_gridcv = GridSearchCV(estimator=rfc_gridcv,  
param_grid=params, cv= 10, n_jobs = -1)  
rfc_gridcv.fit(X_train, y_train)  
print("Best Hyper Parameters:\n",rfc_gridcv.best_params_)
```

Result

Best Hyper Parameters:

```
{'criterion': 'entropy', 'max_depth': 10, 'n_estimators': 500}
```

Model and The DM algorithms – SVM

- Although SVM algorithm can be effective in high-dimensional spaces and have different kernel functions, the training time can be lengthy with large datasets.

```
param_grid = {  
    'C': [0.1, 1, 10],  
    'kernel': ['rbf', 'poly'],  
    'gamma': ['scale', 'auto'],  
}
```

```
svm_classifier = SVC(random_state=42)  
grid_search = GridSearchCV(svm_classifier, param_grid, cv=10)  
grid_search.fit(X_train, y_train)
```

Best Hyper Parameters:

```
{'C': 1, 'gamma': 'scale', 'kernel': 'rbf'}
```

Result

Model and The DM algorithms - XGBoost

- XGBoost belongs to a category of the ensemble learning. It is effective for structured data and has high performance compared to other algorithms.

```
xgb_params = {
    'learning_rate': [0.1, 0.2],
    'n_estimators': [100, 200, 300, 400, 500],
    'max_depth': [3, 5, 7],
    'colsample_bytree': [0.8, 1.0],
    'gamma': [0, 1, 5],
}

xgb_classifier = xgb.XGBClassifier(random_state=42)
xgb_grid_cv = GridSearchCV(xgb_classifier, param_grid=xgb_params, cv=10)
xgb_grid_cv.fit(X_train, y_train)
print("Best Hyper Parameters:\n",xgb_grid_cv.best_params_)

Best Hyper Parameters:
{'colsample_bytree': 0.8, 'gamma': 0, 'learning_rate': 0.1, 'max_depth': 3,
'n_estimators': 300}
```

Result

05



Performance Evaluation

Used Performance Measurements

Used Performance Measurements

- To acquire more accuracy rather than obtaining from training, **10-fold cross-validation** is implemented.
- In addition, **sensitivity**, **specificity**, and **F1 score** are also examined.

Sensitivity - identify positive instances

Specificity – identify negative instances

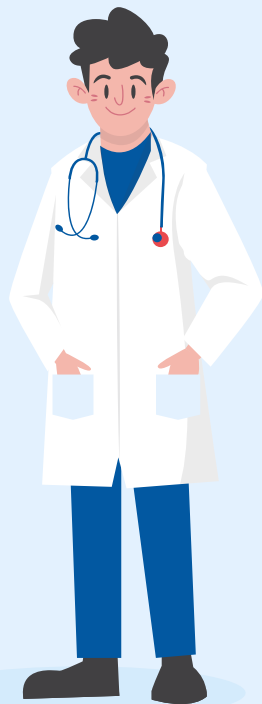
F1 Score - a balance between false positives and false negatives

		Actual Values	
		Positive	Negative
Predicted Values	Positive	Numbers of Cases Predicted with CVD and Actually had CVD (TP)	Numbers of Cases Predicted with CVD and Actually had No CVD (FP)
	Negative	Numbers of Cases Predicted with No CVD and Actually had CVD (FN)	Numbers of Cases Predicted with No CVD and Actually had No CVD (TN)
		Sensitivity/ Recall	Specificity

Used Performance Measurements

	K-NN	Random Forest	SVM	XGBoost
Accuracy	0.71	0.7226	0.7277	0.7293
Sensitivity	0.68	0.68	0.67	0.70
Specificity	0.75	0.76	0.78	0.75
F1 Score	0.7	0.71	0.71	0.72

06



Conclusions & Discussions

The Discovered Knowledge
Future Works

Conclusions & Discussions

- In conclusion, from data preprocessing stage, we can know the relationship/factor that can lead to CVDs, e.g., 'age', 'ap_hi', 'ap_lo', and 'cholesterol'.
- And utilizing these attributes to construct different classification models.
- However, BMI factor is less correlation with target value compared to above factors, which surprise me.
- In the comparison of three machine learning algorithms, **XGBoost** outperform slightly than other three methods in general.

Conclusions & Discussions

- The accuracy of three algorithms are very closely and none of them can achieve over 0.8%. This situation may cause by the low correlation between those input features.
- In the furture, some techniques can be taken to enhance the models' performance, e.g., correlated data, more clean data, utilizing other algorithms, etc.

Reference

Cardiovascular diseases. https://www.who.int/health-topics/cardiovascular-diseases#tab=tab_1

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Ulianova S. (2019) Cardiovascular disease dataset. <https://www.kaggle.com/sulianova/cardiovascular-disease-dataset>

郭緒東, 周宜興, 張麟生, & 張天長. (2011). 心血管疾病於飲食及生活型態方面的初級預防. 家庭醫學與基層醫療, 26(7), 301-307.

111年國人死因統計結果. <https://www.mohw.gov.tw/cp-16-74869-1.html>

Thank you for listening!

Any questions?

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Supplementary Information

- **Descriptive Analysis**

第2單元 描述統計 | 心理科學基礎統計 (scgeeker.github.io)

尺度的類型 — 名目尺度、順序尺度、區間尺度、比例尺度 - Marketingdatascience – Medium

資料分析前，必須知道的四種數據類型 (aif.tw)

資料分析學習筆記 - 資料偏態處理. 相信無論是在學統計，或是進階到機器學習，都會常常聽到「在假設資料常態分佈的情況下... | by Jason | Medium

- **Data Preprocessing**

[資料分析&機器學習] 第2.4講：資料前處理(Missing data, One-hot encoding, Feature Scaling) - JamesLearningNote – Medium

【Python】機器學習 — 資料前處理與樹模型 | by NTU Data Analytics Club | Medium

机器学习之数据的偏态分布和数据的标准化的偏态分布的数据怎么分析-CSDN博客

[Day 27] 機器學習常犯錯的十件事 - iT 邦幫忙::一起幫忙解決難題，拯救 IT 人的一天 (ithome.com.tw)

